



RECOMMENDED PRACTICE: **OFF-ROADWAY SIGN LUMINANCE** AN AMERICAN NATIONAL STANDARD



ANSI/IES RP-39-19

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OFF-ROADWAY SIGN LUMINANCE**
AN AMERICAN NATIONAL STANDARD

Publication of this Technical Memorandum
has been approved by IES.
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should be directed to IES.

Prepared for IES
By the Outdoor Environmental Lighting Committee



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1.0 Introduction and Scope

1.1 Introduction

The effect on the community and environment should be carefully weighed whenever lighting of off-roadway signs is considered. Lighting should maintain the minimum luminance required for visibility and be extinguished when no longer needed, to minimize the negative impact of glare, light trespass, sky glow, animal attraction, and driver distraction. The recommendations in this document include restrictions on maximum sign luminance; lower luminance may be desirable and appropriate, depending on the graphic content of the sign and the background luminance of the surroundings. The design of the graphic content of the sign contributes prominently to the visibility of the sign and should be considered. In general, light backgrounds and dark text produce more glare, light trespass, and sky glow than dark backgrounds and light text.²

In developing the recommendations included in this document, the transient adaption of the human eye has been a primary focus and concern. The human eye is only capable of handling a limited range of luminances at any given time, spanning a range of two to four log units.³ This is the range for adaptation via synaptic interaction, a very fast process. A wider range requires additional adaptation on the part of the visual system, such as pupil size change and, eventually, pigment bleaching, which happen more slowly and can take longer to recover from. While the adaption to a brighter light source happens relatively quickly, adapting to a lower lighting level takes much longer. During this time, visual acuity is diminished; therefore, rapid changes in contrast ratios should be avoided.⁴ In addition, the time required to readapt to lower illuminance levels increases as the eye ages.⁵

1.2 Scope

These recommendations provide guidelines only for the lighting of signs that are located off the right of way of roadways—in other words, all signs not regulated by a federal, state, provincial, or local jurisdiction—and includes on- and off-premise, internally and externally illuminated, and electronic signs. For recommendations for roadway signs used for vehicle or pedestrian

navigation, the reader is referred to *ASNI/IES RP-8-18, Recommended Practice for Design and Maintenance of Roadway and Parking Facility Lighting*.¹

2.0 Definitions

animated sign: A sign employing actual motion, the illusion of motion, or light and/or color changes achieved through mechanical, electrical, or electronic means.

billboard: See *outdoor advertising sign*.

building sign: A sign that is applied or affixed to a building.

candela per square meter (cd/m²): The SI (metric) unit used to describe the luminance of a light source or of an illuminated surface that reflects or transmits light. Also referred to as a *nit*.

changeable sign: A sign with the capability of content change by means of manual or remote input, includes the following types:

- manually activated: The message copy or content can be changed manually on a display surface.
- electrically activated: The message copy or content can be changed by means of remote, electrically energized, on-off switching combinations of alphabetic or pictographic components arranged on a display surface. Illumination may be integral to the components, such as characterized by lamps or other light-emitting devices, or it may be from an external light source designed to reflect off the changeable component display.

commercial outdoor advertising sign: A permanent off-premise sign erected, maintained or used in the outdoor environment for the purpose of providing copy area for commercial or noncommercial messages.

electric sign: Any sign activated or illuminated by means of electrical energy.

electronic message center or electronic sign (EMC):

An electrically activated changeable sign whose variable message and/or graphic presentation capability can be electronically programmed by computer from a remote location. EMCs typically use light emitting diodes (LEDs) as a lighting source.

exterior sign: Any sign placed outside a building.

externally illuminated sign: See *illuminated sign*.

fully shielded: A method of controlling the directionality of the light emitted by a lighting fixture such that all of the light is projected below the horizontal as determined by photometric test or certified by the manufacturer.

illuminance: The amount of light falling upon a real or imaginary surface, commonly called “light level.” Measured in lux (lumens per square meter, lm/m^2) or footcandles (lumens per square foot, lm/ft^2).

illuminated sign: A sign characterized by the use of electric light, either projecting through its surface(s) (internally illuminated or trans-illuminated) or reflecting off its surface(s) (externally illuminated).

internally illuminated sign: See *illuminated sign*.

luminance: The light that is emitted by or reflected from a surface. Measured in units of luminous intensity per unit area (candelas per square meter, cd/m^2). Sometimes also expressed as “nits,” a colloquial reference to cd/m^2 . Can be measured with a luminance meter.

nit: See *candela per square meter (cd/m^2)*.

off-premise sign: A permanent sign erected, maintained or used in the outdoor environment for the purpose of the display of commercial or noncommercial messages not appurtenant to the use of products sold on, or the sale or lease of, the property on which it is displayed. May also be referred to as a billboard or outdoor advertising sign.

on-premise sign: A sign erected, maintained or used in the outdoor environment for the purpose of the display of

messages appurtenant to the use of, products sold on, or the sale or lease of, the property on which it is displayed.

outdoor advertising sign: See *on-premise sign* and *off-premise sign*.

off-roadway sign: Any sign located off of a roadway and not intended for motorist or pedestrian navigational information. These include all signage not covered by the *ANSI/IES RP-8-18, Recommended Practice for Design and Maintenance of Roadway and Parking Facility Lighting*.

portable sign: Any sign not permanently attached to the ground or building that can be set up and removed without the use of tools. Power to operate the sign shall be by cord connection, or solar or battery source.

SI (International System of Units): The modern metric system of measurement; abbreviated *SI* for the French term *Système International d’Unités*.

sign: Any device visible from a public place and whose essential purpose and design is to convey either commercial or noncommercial messages by means of graphic presentation of alphabetic or pictorial symbols or representations. A municipal, state, provincial, or federal flag displayed from a flagpole or staff shall not be considered a sign.

sign copy: The letters, numerals, figures, symbols, logos and graphic elements comprising the content or message of a sign, exclusive of numerals identifying a street address only.

temporary sign: A sign intended to display either commercial or noncommercial messages of a transitory or temporary nature. A portable sign, or any sign not permanently embedded in the ground or not permanently affixed to a building or sign structure that is permanently embedded in the ground, is considered a temporary sign.

transient adaptation: A special type of adaptation, where the human eye has to adapt from low to high light levels and back in short intervals. This happens when the visual environment has very high *contrasts*;

e.g., a computer monitor ($< 200 \text{ cd/m}^2$) and a sunlit wall outside a window ($> 5,000 \text{ cd/m}^2$) can be seen next to each other without turning the head. Excessive transient adaptation soon results in eye fatigue.

transition: A visual effect used on an electronic message sign to change from one message to another.

window sign: A sign affixed to the surface of a window with its message intended to be visible to the exterior environment.

3.0 Lighting Zones

Similar to land use zoning, lighting zones are set by a community to help define desired ambient light levels. The lighting zones range from an area or district with no ambient light (LZ0) to one of very high ambient light (LZ4). Default zones will typically be LZ1, LZ2 or LZ3.

Lighting zones have the following attributes:

- Help communities minimize the contrast (and conflict) between extremes in lighting, such as a brightly lighted car dealership adjacent to or within line of sight of a residential neighborhood
- May determine restrictions on outdoor lighting that impacts “places of sleep,” such as residential areas, hospitals, and long-term care facilities
- May employ vertical distinctions, such as in mixed-use facilities where the commercial aspects are at street level and residential units are on the higher levels
- Encourage minimal changes in visual adaptation when traveling from site to site

Brief descriptions follow. For more details on the determination and use of lighting zones, the reader is referred to the *IDA/IES Model Lighting Ordinance*,⁶ or *IES RP-33-14, Lighting for Exterior Environments*.⁷

LZ0: No ambient light. Areas where the natural environment could be affected seriously and adversely by small amounts of electric lighting at night. This is the recommended default zone for wilderness areas, parks and preserves, and undeveloped rural areas.

LZ1: Low ambient light. Developed areas within a natural environment and areas of human activity that are inherently dark at night. This is the recommended default zone for rural and low-density residential areas.

LZ2: Moderate ambient light. Areas of human activity (i.e., habitation, recreation and/or work) where electric lighting *may* be required for safety and convenience at night. This is the recommended default zone for light-commercial business districts and high-density or mixed-use residential districts.

LZ3: Moderately high ambient light. Areas of human activity (i.e., habitation, recreation and/or work) where electric lighting may be continuous and is required for safety and convenience at night. This is the recommended default zone for large cities’ business districts.

LZ4: Very high ambient light. Areas of very high levels of human activity at night, including significant interaction among pedestrians and/or vehicles. Most cities do not have areas that meet this criterion. This is not a default zone.

4.0 Recommendations

4.1 Sign Luminance Recommendations

A sign may not exceed a maximum luminance level in any direction, measured in candelas per square meter (cd/m^2), or nits, based on the lighting zone (see **Section 3.0**) in which it is located, regardless of the method of illumination. These maximum luminance levels^{8,9,10} are provided in **Table 4-1**.

Table 4-1. Maximum Sign Luminance by Lighting Zone (cd/m^2 , or nits)

LIGHTING ZONE	NIGHTTIME	DAYTIME
LZ0	0	0
LZ1	20	3,500
LZ2	40	3,500
LZ3	80	3,500
LZ4	160	3,500

These recommendations are the same for any technology used to produce sign lighting. Many measures of performance and perception have been shown to depend on luminance, because it is one of the direct stimuli to vision. For this reason, all recommendations are made using cd/m^2 or nits, and a calibrated luminance meter is required to accurately measure sign lighting for compliance. (See **Section 4.4 Measurement of Maximum Sign Luminance**.)

All illuminated signs shall comply with the maximum luminance levels specified in **Table 4-1** for nighttime conditions beginning at least one-half hour before apparent sunset for the specific geographic location and date. All illuminated signs shall comply with this maximum luminance level throughout the night, if the sign is energized, until no later than one-half hour before apparent sunrise, at which time the sign may resume luminance levels appropriate for daytime conditions, when required or appropriate.

4.2 Environmental Considerations

Signs that have external illumination, whether the light source is mounted above or below the sign face or panel, shall have lighting fixtures (luminaires) that are aimed toward the sign surface. The luminaire optical system shall restrict the illumination to the sign face and minimize light trespass and off-site glare.

4.3 Electronic Sign Message Transition and Duration Recommendations

Message duration should be limited to not less than twenty seconds in length, and no motion video or rapidly flashing graphic elements are permitted.^{11,12,13}

4.4 Measurement of Maximum Sign Luminance

Measurements of maximum sign luminance shall follow the following procedures:

- Sign luminance shall be measured using a luminance meter with a current NIST^{*}-traceable calibration. The use of a luminance meter with a $\frac{1}{3}$ -degree field of view is recommended.
- Luminance measurements shall be taken during the hours between the end of evening civil twilight and the beginning of morning civil twilight.
- Luminance measurements shall be taken from a position as close to the sign being measured as reasonably possible.
- The light meter (photometer) shall be positioned for measurements such that the portion of the sign being measured fills the central circle of the viewfinder (the “measurement area”). All measurements shall be taken of the sign surface at its maximum level of illuminance. In addition, the field of view of the meter used shall be fully covered by the white or lightest-colored surface of the sign. It may be necessary to move the photometer closer to the sign until this is possible.
- At least 5 individual measurements shall be made for each sign, and the measurements should be averaged.
- The use of a tripod is recommended.
- Measurements shall be taken on cloudless nights with no atmospheric elements such as rain, snow, fog, or other atmospheric particulates that could affect the measurements.

^{*}U.S. National Institute of Science and Technology

References

- 1 Illuminating Engineering Society. ANSI/IES RP-8-18, Recommended Practice for Design and Maintenance of Roadway and Parking Facility Lighting. New York: IES; 2018.
- 2 Terry T, Gibbons R. National Surface Transportation Safety Center for Excellence (NSTSCE). Assessment of the Impact of Color Contrast in the Detection and Recognition of Objects in a Road Environment. Blacksburg, VA.: Virginia Tech Transportation Institute; 2011 Dec 16. (NTSTSCE 11-UL-016).
- 3 DiLaura D, Houser K, Mistrick R, Steffy G (editors). The Lighting Handbook, 10th ed. New York: Illuminating Engineering Society; 2011.
- 4 Lamb TD, Pugh EN Jr. Phototransduction, Dark Adaptation, and Rhodopsin Regeneration. *Investigative Ophthalmol Vis Sci.* 2006;47:5138–52.
- 5 Jackson GR, Owsley C, McGwin G Jr. Aging and dark adaptation. *Vision Res.* 1999;39(23):3975–82.
- 6 Illuminating Engineering Society (IES) and International Dark-sky Association (IDA). Joint IDA-IES Model Lighting Ordinance (MLO) with User Guide. New York: IES; 2011.
- 7 Illuminating Engineering Society. IES RP-33-14, Lighting for Exterior Environments. New York: IES; 2014.
- 8 Guterman PS, Fukuda K, Wilcox LM, Allison RS. Is Brighter Always Better? The Effects of Display and Ambient Luminance on Preferences for Digital Signage. *SID 10 Digest.* 2010;41(1);1116–19.
- 9 Freyssinier JP, Narendran N, Bullough J. Luminance requirements for lighted signage. In: *Proceedings Sixth Internatl Conf on Solid State Lighting*, San Diego, CA.; 2006; doi: 10.1117/12.681422. (63371M).
- 10 Luginbuhl CB, Israel H, Scowen P, Polakis J, Polakis T. Digital LED Billboard Luminance Recommendations – How Bright is Bright Enough? Flagstaff, AZ.: U.S. Naval Observatory Flagstaff Station; 2010.
- 11 Wachtel J, Netherton R. Safety and Environmental Design Considerations in the Use of Commercial Electronic Variable-Message Signage. Washington, DC: Federal Highway Administration. 1980. (FHWA-RD-80-051).
- 12 Wisconsin Department of Transportation District 2, Freeway Operations Unit. Milwaukee county stadium variable message sign study: Impacts of an advertising variable message sign on freeway traffic. Madison, WI.: WDOT; 1994.
- 13 Divekar G, Pradhan A, Pollatsek A, Fisher D. Effect of External Distractions Behavior and Vehicle Control of Novice and Experienced Drivers Evaluated. *Transp Res Record J Transp Res Board.* 2013;2321:15–22.

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Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

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The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

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